

PROFESSIONAL SUMMARY

Computer Science Graduate specializing in **Embedded Systems & IoT (ESP32)**, **Computer Vision (OpenCV)**, and **Full-Stack Web Development**. Experienced in designing distributed multi-node telemetry networks, automated license plate and facial recognition pipelines, and scalable web platforms with Vue 3 and Python Flask. Passionate about engineering intelligent physical and software systems that solve tangible, real-world problems.

TECHNICAL SKILLS

Embedded Systems & IoT: ESP32, C/C++, Arduino, PlatformIO, ThingsBoard Cloud, MQTT Telemetry, Environmental Sensor Networks
Computer Vision & AI: OpenCV, Python, CNNs, Automatic License Plate Recognition (ALPR), Facial Recognition, Scikit-learn, TensorFlow
Web & Database Engineering: Vue.js 3, TypeScript, JavaScript, Python Flask, RESTful APIs, MySQL, HTML5 / CSS3
Tools & Workflow: Git, GitHub, Linux OS, VS Code, PlatformIO, Vercel

PROFESSIONAL EXPERIENCE

Machine Learning Developer • **Oops Automate**10/2025 – 12/2025 | Wah Cantt

- Architected and deployed an AI-powered conversational assistant for Shopify handling real-time customer queries, product discovery, and semantic search.
- Engineered an AI chatbot for eBay integrating live eBay REST APIs to fetch real-time listings, price comparisons, and intelligent recommendations.
- Optimized LLM response latency and conversational context retention for production e-commerce workflows.

Internship Trainee (Data Engineering & ML) • **Bytewise Limited**06/2024 – 09/2024 | Remote

- Processed raw data utilizing Python, SQL, and Pandas—resolving missing values, outliers, and schema inconsistencies.
- Engineered automated data preprocessing pipelines and feature transformations for predictive machine learning workflows.

FEATURED PROJECTS

ESP32 Multi-Module Environmental SystemGitHub ↗
C++, ESP32, PlatformIO, ThingsBoard Cloud, IoT Telemetry

- Engineered a 3-node distributed IoT monitoring network tracking air quality (MQ-137, CCS811), water purity (TDS & Flow), and soil/light metrics (BH1750).
- Implemented MQTT telemetry publishing to ThingsBoard Cloud dashboards for real-time visualization and automated threshold alerts.

Vehicle Detection & ALPR (Automatic License Plate Recognition)GitHub ↗
Python, OpenCV, Computer Vision, Machine Learning

- Developed an automated computer vision pipeline executing real-time vehicle detection and license plate extraction from traffic surveillance feeds.
- Applied image morphological transformations and character segmentation algorithms to achieve high recognition accuracy in variable conditions.

Face Attendance Automation SystemGitHub ↗
Python, OpenCV, Facial Recognition, Biometrics

- Built a real-time biometric attendance system utilizing 128-dimensional facial encodings, live webcam video streams, and automated attendee logging.

My-Bakery E-Commerce PlatformLive Demo ↗
Vue.js 3, JavaScript, CSS3, Vercel

- Designed and deployed a responsive online retail storefront featuring dynamic product catalog filtering, cart state management, and real-time UI.

AdvanHR – AI HR Assistant (Oman Government Research Project)Research Project
Python, LLMs, NLP, Document Processing

- Automated routine HR queries, employee turnover forecasting, and document processing via an intelligent virtual assistant powered by LLMs.

EDUCATION

BS in Computer Science11/2021 – 06/2025
UET Taxila • Pakistan

CERTIFICATIONS & LANGUAGES

- Supervised Machine Learning (Stanford / DeepLearning.AI)
- Azure AI Fundamentals & Azure Cloud Fundamentals
- Google: Crash Course in Python
- Languages:** English (Professional), Urdu (Native)